

Graph-conditioned state-space models for genetic perturbation response prediction: a controlled evaluation against simple baselines

Aryan Padarthy^{1,*} and Riley Huynh¹

¹Allen High School, Allen, TX 75013, USA

*To whom correspondence should be addressed.

Associate Editor: XXXXXXXX

Received on XXXXX; revised on XXXXX; accepted on XXXXX

Abstract

Motivation: Predicting the transcriptional response to unmeasured genetic perturbations—in particular to combinations of perturbations—is a central goal of single-cell perturbation modelling, yet recent benchmarks show that deep models rarely beat simple additive or linear baselines. State-space models (SSMs) scale linearly with the number of genes, but genes have no natural order and an SSM has no notion of which genes are related. We asked whether conditioning an SSM on a gene–gene co-expression graph closes this gap.

Results: We introduce GCP-Mamba, a bidirectional selective SSM over all 5,045 measured genes in which the diffusion of the perturbation over a control-cell co-expression graph modulates the discretization step of every gene token. On the five standard GEARS splits of the Norman et al. CRISPRa screen, GCP-Mamba gave the lowest error of all methods on unseen single perturbations, and the true graph outperformed both a label-permuted graph and no graph there. For double perturbations, however, the additive sum of single-perturbation effects remained the best predictor, no model explained any of the variance of genetic interactions beyond it, and graph conditioning of the step size itself had no measurable effect. We also document two evaluation pitfalls: a perturbation encoding restricted to highly variable genes that silently discarded 80% of training perturbations in an earlier version of this work, and a genetic-interaction correlation metric that scores a predictor of no change close to trained models.

Availability and implementation: Code, splits and all per-condition results are available at <https://github.com/The-MathKing/gcpmamba> (MIT license).

Contact: aryan.padarthy@student.allenisd.org

Supplementary information: Supplementary data are available at *Bioinformatics* online.

1 Introduction

Perturb-seq screens measure the transcriptome-wide response of single cells to CRISPR-mediated gene perturbations [Dixit et al., 2016, Norman et al., 2019, Replogle et al., 2022]. Because the number of possible combinations grows quadratically with the number of targets, only a small fraction of double perturbations can ever be measured: the Norman et al. K562 CRISPRa screen profiled 131 of the $\binom{105}{2} = 5,460$ pairs of its 105 targets [Norman et al., 2019]. Models that predict the response to unmeasured perturbations—and in particular the non-additive, genetic-interaction (GI) component of double perturbations—are therefore a central goal of the field.

Knowledge-graph neural networks such as GEARS [Roohani et al., 2024] and TxPert [Wenkel et al., 2026], and transformer-based single-cell foundation models [Cui et al., 2024, Theodoris et al., 2023], have been proposed for this task. However, recent benchmarks found that such models often do not outperform deliberately simple baselines: the

mean training response, the sum of single-perturbation effects, or a low-rank linear model [Ahlmann-Eltze et al., 2025, Csendes et al., 2025]. Two lessons from this literature shape our study: every new architecture must be compared against these baselines on identical splits, and the contribution of any biological prior must be isolated by controls that remove it while keeping everything else fixed.

State-space models (SSMs) such as Mamba [Gu and Dao, 2023] process sequences in time and memory linear in their length, whereas self-attention is quadratic [Vaswani et al., 2017]. Treating the transcriptome as a sequence of gene tokens, an SSM can process all measured genes at once, but genes have no natural order and the recurrence has no notion of which genes are functionally related. We asked whether both limitations can be addressed by conditioning the SSM’s discretization step on a gene–gene graph, and—more importantly—whether doing so yields better predictions than simple baselines.

Our contributions are: (i) GCP-Mamba, a bidirectional selective SSM over all measured genes in which the diffusion of the perturbation over a co-expression graph modulates the step size of every gene token; (ii) a controlled benchmark on the five standard GEARS splits of the Norman

data and on the Adamson et al. CRISPRi screen, against five simple baselines, GEARS retrained on the same splits, and eight ablations and variants that remove, permute or relocate the graph, destroy the gene ordering, remove the recurrence or add an additive prior; and (iii) two evaluation pitfalls, one of which invalidated the conclusions of an earlier version of this work.

2 Methods

2.1 Data and splits

We used the Norman et al. K562 CRISPRa Perturb-seq data [Norman et al., 2019] as distributed with GEARS: 91,205 cells, 5,045 genes, 7,353 control cells, single perturbations of 105 genes and 131 double perturbations, log-normalised. For every condition p we computed the pseudobulk response $\delta_p = \bar{x}_p - \bar{x}_{\text{ctrl}} \in \mathbb{R}^{5045}$. Singles are recorded as two conditions (A+ctrl, ctrl+A) that use different guides; wherever a single response is needed we average them. We used the GEARS *simulation* splits with seeds 1–5, generated with the official `cell-gears` code [Roohani et al., 2024]. Each holds out about a quarter of the targets; test perturbations are grouped into unseen singles and doubles with zero, one or two targets seen during training (“0/2”, “1/2”, “2/2 seen”; GEARS counts a target as seen if it appears in the training or validation set). Across the five splits the test sets contain 185 unseen-single, 32 “0/2”, 254 “1/2” and 90 “2/2” perturbations. Validation perturbations were used only for early stopping and for choosing the ridge penalty of the linear baselines. As a second dataset we used the Adamson et al. K562 CRISPRi screen [Adamson et al., 2016] (86 single perturbations, 5,060 genes; GEARS simulation splits, seeds 1–5).

The co-expression graph, gene embeddings and gene order below were computed from unperturbed control cells only; they are identical in every split and contain no information about any perturbation response.

2.2 Co-expression graph and perturbation signal

We z-scored control-cell expression, computed absolute Pearson correlations between all gene pairs, kept the $k = 20$ strongest neighbours of every gene and symmetrised the result, giving a sparse graph $\mathbf{W}$ with 111,078 edges. For a perturbation set P with indicator $\mathbf{1}_P \in \{0, 1\}^G$, the graph signal is a truncated diffusion

$$c(P) = \log\left(1 + s \sum_{k=1}^3 \beta^{k-1} \widetilde{\mathbf{W}}^k \mathbf{1}_P\right), \quad \widetilde{\mathbf{W}} = \mathbf{D}^{-1/2} \mathbf{W} \mathbf{D}^{-1/2}, \quad (1)$$

with degree matrix $\mathbf{D}$, $\beta = 0.5$ and $s = 10^3$ (the logarithm compresses values that span four orders of magnitude). $c_i(P)$ is large for genes close to a perturbed gene in the graph and zero for genes more than three steps away.

2.3 GCP-Mamba

The model is summarised in Figure 1.

Tokens. Each of the G genes is one token. Token i is the sum of a learned identity vector, a projection of the gene’s control-cell embedding $\mathbf{e}_i$ (its loadings on the top 64 principal components of control-cell expression, standardised) and of its control mean expression. A perturbation enters in three ways: (i) a learned flag vector added to the tokens of the targeted genes; all targets are measured genes, so no perturbation is lost (Section 3.1); (ii) a set embedding $\mathbf{z}_P = \sum_{g \in P} (f(\mathbf{e}_g) + \mathbf{t}_g)$ added to every token, where f is a two-layer MLP and $\mathbf{t}_g$ a per-target vector initialised at zero; only targets perturbed in training receive gradient for $\mathbf{t}_g$, so a target never seen in training is represented through $f(\mathbf{e}_g)$ alone, which is what permits zero-shot

prediction; and (iii) the graph signal $c_i(P)$, linearly projected and added to token i .

Graph-conditioned selective scan. Each of two layers applies a forward and a backward Mamba-style selective SSM [Gu and Dao, 2023] with diagonal state matrix $\mathbf{A} = -\text{diag}(1, \dots, N)$, input-dependent $\mathbf{B}_i$ and $\mathbf{C}_i$, a SiLU output gate and a skip connection. The step size is conditioned on the graph,

$$\Delta_i = \text{softplus}(\mathbf{W}_\Delta \mathbf{u}_i + \mathbf{b}_\Delta + \mathbf{w}_c c_i(P)), \quad (2)$$

where $\mathbf{u}_i$ is the layer input and $\mathbf{w}_c \in \mathbb{R}^d$ is learned and initialised at zero, so that the model starts as an unconditioned SSM. With zero-order-hold discretization, $\mathbf{h}_i = \exp(\Delta_i \mathbf{A}) \mathbf{h}_{i-1} + \Delta_i \mathbf{B}_i \mathbf{u}_i$; tokens near the perturbation take larger steps, writing more of their input into the state and forgetting more of the past. Because $\mathbf{w}_c c_i(P)$ differs between genes and between perturbations, permuting the graph changes the function the model computes.

Gene order. Genes are placed along the Fiedler vector [Fiedler, 1973] of the normalised Laplacian of $\mathbf{W}$, so that strongly co-expressed genes are close in the sequence.

Readout. The prediction for gene i is $\hat{\delta}_i = b_i + \mathbf{w}^\top \mathbf{h}_i + (\mathbf{U} \mathbf{h}_i)^\top (\mathbf{V} \mathbf{z}_P)$, where $\mathbf{h}_i$ is the final token state and $\mathbf{U}, \mathbf{V}$ have rank 16. The bilinear term expresses a gene-by-perturbation interaction and contains the low-rank linear model of Ahlmann-Eltze et al. [2025] as a special case. The bias b_i is initialised at the mean training response and $\mathbf{w}$ and $\mathbf{V}$ at zero, so that training starts from the *train-mean* baseline and learns deviations from it; without this initialisation, and without standardising $\mathbf{e}_g$, the model did not improve on the train-mean baseline in pilot runs on the validation set of split 1 (Supplementary Section S2).

Implementation and complexity. We implemented the scan in PyTorch as an exact chunked parallel scan: cumulative sums in log space within chunks of 16 tokens and a sequential recurrence across chunk states. Per-step log-decays are clamped at -5 (a decay factor of at least 0.0067) so that the within-chunk exponentials remain finite in single precision; the implementation matches a naive sequential recurrence to 10^{-5} (unit test in the repository). Time and memory are $\mathcal{O}(BGdN)$ for the scan and $\mathcal{O}(|\mathbf{W}|)$ for the sparse diffusion, i.e. linear in the number of genes; no $G \times G$ object is formed.

Training. $d = 32$, $N = 4$, dropout 0.1 in f ; AdamW [Loshchilov and Hutter, 2019] with learning rate 10^{-3} , weight decay 10^{-2} and batch size 8; gradient clipping at 1; mean-squared error over all genes. Models were trained for at least 40 and at most 100 epochs and stopped 15 epochs after the last improvement of the validation loss; the state with the lowest validation loss was kept. (The validation loss typically plateaus for several epochs before a model starts to use the perturbation; without the 40-epoch minimum, patience-based stopping truncated some runs during this plateau.) The model seed equals the split seed. All runs used a 4-core CPU; one training run took 5–25 minutes.

2.4 Ablations and variants

All variants share the architecture, perturbation inputs (i)–(ii), optimiser and stopping rule, and differ in one factor. Δ *only*: the graph enters only through Δ , not through the tokens. Δ *only*, $\mathbf{w}_c \sim \mathcal{N}(0, 1)$: as before, but $\mathbf{w}_c$ is initialised randomly so that the conditioning is active from the start. *Permuted graph*: the gene labels of $\mathbf{W}$ are randomly permuted, which preserves its degree distribution and spectrum. *Random order*: genes are placed in random order instead of the Fiedler order. *Mamba (no graph)*: $c(P) \equiv 0$. *Graph-MLP*: the SSM layers are replaced by per-token MLPs, so genes interact only through $\mathbf{z}_P$ and the graph signal. *Additive prior*: the network predicts the residual $\delta - \pi$ with respect to the additive prior π (the sum of the targets’ observed single responses, or the mean single

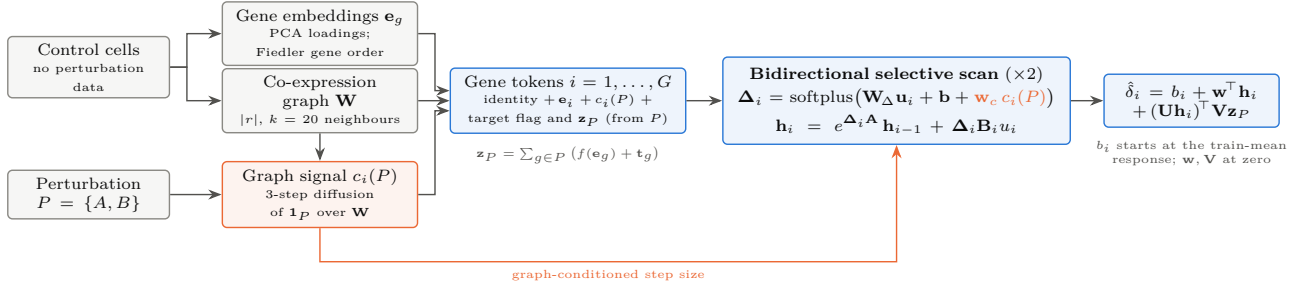


Fig. 1. GCP-Mamba. Everything derived from expression (grey) is computed from control cells only. The perturbation enters the gene tokens (a target flag and the set embedding $\mathbf{z}_P$) and, through its diffusion $c_i(P)$ over the co-expression graph (orange), the step size Δ_i of every token in the bidirectional selective scan. The bilinear readout starts at the train-mean prediction.

response for an unseen target; for single perturbations always the mean single response), with and without the graph.

2.5 Baselines

No change predicts $\delta = \mathbf{0}$. *Train mean* predicts the mean training response. *Additive* predicts the sum of the targets’ single responses observed during training (training or validation set), substituting the mean single response for a target without one. *Linear* is the model of Ahlmann-Eltze et al. [2025], $\mathbf{Y} \approx \mathbf{G}\mathbf{M}\mathbf{P}^\top + \mathbf{b}$, with gene embeddings $\mathbf{G}$ from the top 10 principal components of the training responses and each perturbation represented by the sum of its targets’ embeddings, taken either from $\mathbf{G}$ (*Linear (PCA)*, as in the original) or from the control-cell embeddings e_g that GCP-Mamba receives (*Linear (ctrl emb.)*); the ridge penalty was chosen on the validation set from $\{10^{-3}, \dots, 10^4\}$. *GEARS* was retrained with the official implementation and default hyperparameters (hidden size 64, 20 epochs) on the same splits.

2.6 Metrics and statistics

Following Roohani et al. [2024], for every test perturbation we computed the mean squared error (MSE) and the Pearson correlation between predicted and observed δ on its top-20 differentially expressed (DE) genes. For double perturbations AB we computed the GI residual $\epsilon = \delta_{AB} - \delta_A - \delta_B$ from the measured singles (used only for scoring). The MSE of a predicted residual $\hat{\delta}_{AB} - \delta_A - \delta_B$ equals the MSE of $\hat{\delta}_{AB}$ itself, so it carries no extra information. We therefore report the fraction of GI variance explained, $R_{\text{GI}}^2 = 1 - \sum_{AB} \text{MSE}_{AB} / \sum_{AB} \overline{\epsilon_{AB}^2}$, on “2/2 seen” doubles, where the additive baseline is exactly the measured-singles sum and scores 0, and the Pearson correlation between predicted and observed residuals used in earlier work. Metrics were averaged over the test perturbations of a split and are reported as mean $\pm$ s.d. over the five splits. We compared GCP-Mamba with every other model using two-sided Wilcoxon signed-rank tests on per-perturbation differences pooled across splits, with Holm correction within each metric and test group (Supplementary Table S3). Because perturbations recur across splits these pooled tests are approximate, and we emphasise effect sizes and their consistency across splits.

3 Results

3.1 A perturbation encoding restricted to highly variable genes discards most perturbations

An earlier version of this study encoded a perturbation as a multi-hot vector over the 500 most highly variable genes (HVGs), the same

genes whose expression was predicted, and parsed single perturbations from the condition name with a rule that mapped `ctrl+A` to the control condition. Only 12–17 of the 105 CRISPRa targets were among the HVGs (mean 14 over the ten splits used then), so 77–84% of training perturbations (mean 80%) and 70–94% of test perturbations received an all-zero input, and the 161–183 training perturbations of a split shared only 14–19 distinct input vectors (Supplementary Table S1; `gcpm/audit_hvg_encoding.py`). A model cannot distinguish perturbations it receives as identical inputs, so that version could at best learn the mean response; its negative conclusion about graph conditioning was an artefact of the encoding, not a property of the architecture. The problem is easy to miss: training converges, loss curves look normal, and the model is competitive with the train-mean baseline. It applies to any model whose perturbation input is indexed by the output gene set. Here every target is represented by its own token and by an embedding computed from control cells, and we recommend that perturbation models report how many targets reach the input.

3.2 Unseen single perturbations benefit from the graph; double perturbations do not

Table 1 and Figure 2 compare GCP-Mamba with the baselines and GEARS on the Norman data. As in previous benchmarks [Ahlmann-Eltze et al., 2025], the additive baseline was the most accurate predictor overall (MSE 0.240 on the top-20 DE genes, averaged over all test perturbations) and for doubles with at least one seen target; for “2/2 seen” doubles its error (0.089) was half that of any trained model. GCP-Mamba (0.278) improved substantially on the train-mean (0.428), linear (0.349) and no-change (0.561) baselines, and on GEARS (TBD).

For unseen single perturbations, where no additive information exists, GCP-Mamba had the lowest error of all methods (0.255 vs. 0.270 for the train mean, 0.269 for the linear model and TBD for GEARS; Holm-adjusted $P = 0.039$ vs. train mean and $P = 0.24$ vs. the linear model). On the same perturbations the true co-expression graph outperformed a label-permuted graph (0.257; $P = 0.017$) and no graph (0.266; $P = 0.014$), so the gain on unseen singles depends on the content of the graph and not only on its presence. The effects are small, however: the median per-perturbation improvement over the train mean was 0.003 (1–2% of the error), smaller than the variation between splits (Fig. 2). For doubles with only one seen target (“1/2 seen”) the recurrent models were far behind the additive baseline (0.309 vs. 0.239) and adding the graph to the tokens increased the error relative to no graph (0.309 vs. 0.292; $P = 0.019$).

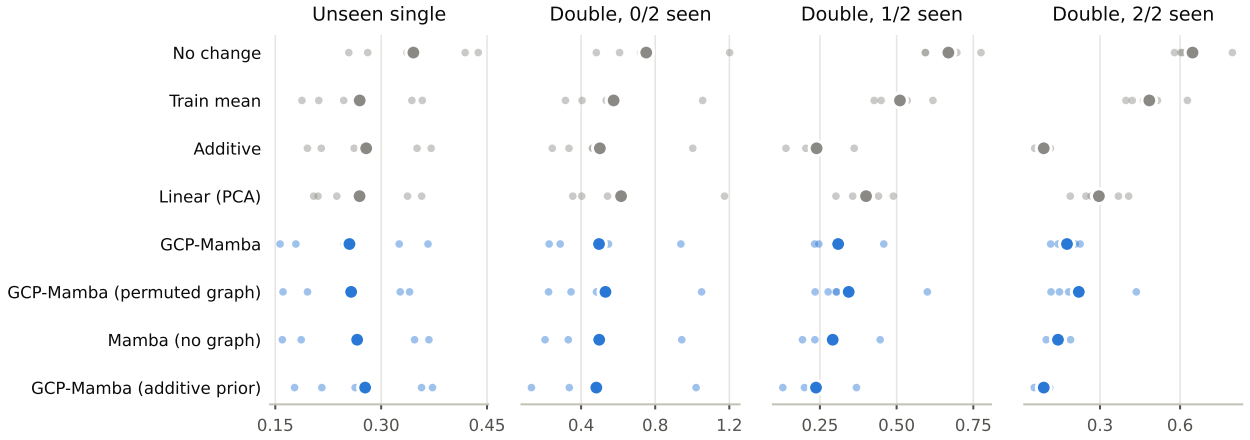
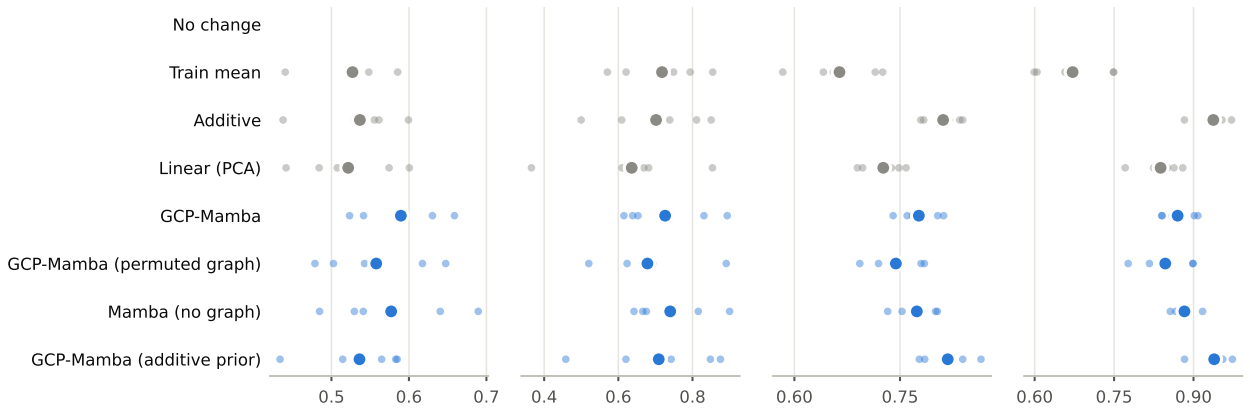
MSE, top-20 DE genes (lower is better)**Pearson r of δ , top-20 DE genes**

Fig. 2. Norman et al. CRISPRa data, five GEARS simulation splits. Each large dot is the mean over splits and each small dot one split; columns are test groups. Top: MSE on the top-20 DE genes of each test perturbation (lower is better). Bottom: Pearson correlation of predicted and observed change on the same genes (undefined for *No change*). Grey, baselines; orange, GEARS; blue, GCP-Mamba and variants.

Table 1. Norman et al. CRISPRa screen: MSE of the predicted expression change on the top-20 DE genes of each test perturbation (mean $\pm$ s.d. over the five GEARS simulation splits; lower is better; best in bold)

Model	Unseen single	Double, 0/2 seen	Double, 1/2 seen	Double, 2/2 seen	All test
No change	0.346 $\pm$ 0.081	0.752 $\pm$ 0.272	0.669 $\pm$ 0.077	0.647 $\pm$ 0.087	0.561 $\pm$ 0.061
Train mean	0.270 $\pm$ 0.077	0.576 $\pm$ 0.287	0.511 $\pm$ 0.076	0.485 $\pm$ 0.092	0.428 $\pm$ 0.061
Additive	0.279 $\pm$ 0.079	0.501 $\pm$ 0.295	0.239 $\pm$ 0.081	0.089 $\pm$ 0.022	0.240 $\pm$ 0.070
Linear (PCA)	0.269 $\pm$ 0.073	0.616 $\pm$ 0.328	0.400 $\pm$ 0.073	0.296 $\pm$ 0.090	0.349 $\pm$ 0.059
Linear (ctrl emb.)	0.322 $\pm$ 0.103	0.962 $\pm$ 0.788	0.518 $\pm$ 0.090	0.413 $\pm$ 0.122	0.452 $\pm$ 0.078
GCP-Mamba	0.255 $\pm$ 0.091	0.497 $\pm$ 0.280	0.309 $\pm$ 0.090	0.176 $\pm$ 0.046	0.278 $\pm$ 0.079

3.3 Graph conditioning of the step size has no measurable effect

Table 2 isolates the components of GCP-Mamba. Conditioning the step size alone (Δ only) gave results indistinguishable from a plain bidirectional Mamba without the graph (0.274 vs. 0.269 overall), and the learned conditioning weights stayed close to their zero initialisation ($\|\mathbf{w}_c\| = 0.07$ per SSM, against 0.56 when the graph is also added to the tokens). This was not merely a failure to move away from zero: initialising $\mathbf{w}_c$ randomly, so that the step size depends strongly on graph proximity from the first update ($\|\mathbf{w}_c\| \approx 5.8$ after training), again changed the

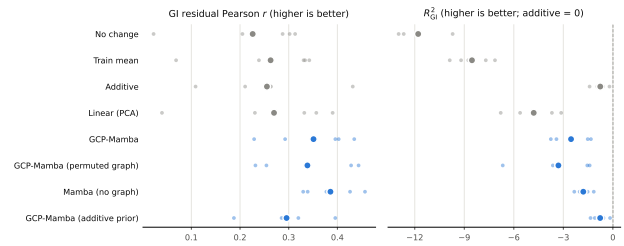
error only marginally (0.276). Graph proximity to the perturbation thus had little influence on how the SSM propagates information, whichever way the conditioning was initialised. What distinguished the models was instead the recurrence: replacing the scan with per-gene MLPs increased the error (0.291 vs. 0.278; $P = 4 \times 10^{-4}$), while the Fiedler ordering was not better than a random order (0.276).

Table 2. Ablations and residual variants on the Norman data: MSE on the top-20 DE genes (mean $\pm$ s.d. over five splits; best in bold)

Model	Unseen single	Double, 0/2 seen	Double, 1/2 seen	Double, 2/2 seen	All test
GCP-Mamba	0.255 $\pm$ 0.091	0.497 $\pm$ 0.280	0.309 $\pm$ 0.090	0.176 $\pm$ 0.046	0.278 $\pm$ 0.079
GCP-Mamba (Δ only)	0.260 $\pm$ 0.086	0.503 $\pm$ 0.297	0.300 $\pm$ 0.116	0.158 $\pm$ 0.064	0.274 $\pm$ 0.095
GCP-Mamba (Δ only, $\mathbf{w}_c \sim \mathcal{N}(0, 1)$)	0.266 $\pm$ 0.091	0.492 $\pm$ 0.296	0.304 $\pm$ 0.088	0.154 $\pm$ 0.024	0.276 $\pm$ 0.081
GCP-Mamba (permuted graph)	0.257 $\pm$ 0.079	0.531 $\pm$ 0.317	0.344 $\pm$ 0.146	0.221 $\pm$ 0.126	0.304 $\pm$ 0.115
GCP-Mamba (random order)	0.263 $\pm$ 0.078	0.526 $\pm$ 0.232	0.303 $\pm$ 0.092	0.158 $\pm$ 0.053	0.276 $\pm$ 0.075
Mamba (no graph)	0.266 $\pm$ 0.093	0.498 $\pm$ 0.279	0.292 $\pm$ 0.096	0.142 $\pm$ 0.034	0.269 $\pm$ 0.086
Graph-MLP (no scan)	0.258 $\pm$ 0.080	0.526 $\pm$ 0.290	0.323 $\pm$ 0.110	0.203 $\pm$ 0.082	0.291 $\pm$ 0.089
GCP-Mamba (additive prior)	0.277 $\pm$ 0.086	0.482 $\pm$ 0.330	0.237 $\pm$ 0.087	0.089 $\pm$ 0.022	0.238 $\pm$ 0.077
Mamba (additive prior)	0.279 $\pm$ 0.085	0.491 $\pm$ 0.328	0.237 $\pm$ 0.088	0.089 $\pm$ 0.022	0.239 $\pm$ 0.076

Table 3. Prediction of genetic interactions (GI) in double perturbations (top-20 DE genes; mean $\pm$ s.d. over five splits). R_{GI}^2 is the fraction of the variance of the GI residual ϵ that a prediction explains; an exactly additive prediction scores 0. The GI Pearson r gives the trivial *No change* and *Train mean* predictors scores close to those of the deep models.

Model	$R_{GI}^2 \uparrow$	GI Pearson $r \uparrow$
No change	-11.79 $\pm$ 1.28	0.23 $\pm$ 0.12
Train mean	-8.53 $\pm$ 1.11	0.26 $\pm$ 0.12
Additive	-0.76 $\pm$ 0.45	0.26 $\pm$ 0.12
Linear (PCA)	-4.79 $\pm$ 1.47	0.27 $\pm$ 0.14
Linear (ctrl emb.)	-7.03 $\pm$ 1.66	0.27 $\pm$ 0.11
GCP-Mamba	-2.53 $\pm$ 1.09	0.35 $\pm$ 0.09
Mamba (no graph)	-1.79 $\pm$ 0.46	0.39 $\pm$ 0.05
GCP-Mamba (additive prior)	-0.77 $\pm$ 0.46	0.30 $\pm$ 0.07
Mamba (additive prior)	-0.76 $\pm$ 0.46	0.27 $\pm$ 0.11

**Fig. 3.** Genetic-interaction prediction on “2/2 seen” doubles of the Norman data. Left: Pearson correlation between predicted and observed GI residuals, which rates trivial predictors close to trained models. Right: fraction of GI variance explained (R_{GI}^2 ; the additive baseline is 0 by construction). Dots as in Fig. 2.

3.6 Computational cost

Figure ?? reports the measured time and peak memory of one training step as a function of the number of gene tokens. TBD

3.4 No model predicts genetic interactions beyond the additive baseline

Double perturbations are interesting chiefly for their non-additive component. Letting the network predict the residual with respect to the additive prior closed the gap to the additive baseline overall (0.238 vs. 0.240) but did not improve on it (Table 2), and on “2/2 seen” doubles no trained model explained any of the GI variance: R_{GI}^2 was essentially zero for the residual models (i.e. they learned a negligible correction to the additive prior) and strongly negative for all others (Table 3).

The GI Pearson correlation, used to assess GI prediction in earlier work including the first version of this study, told a different story (Fig. 3). It rated the recurrent models above the additive baseline, and it gave the trivial no-change and train-mean predictors values (0.23 and 0.26) comparable to the additive baseline and the linear model. This happens because predicting no change yields a predicted residual of $-(\delta_A + \delta_B)$, which is correlated with the true residual whenever the interaction is buffering (sub-additive), as it often is. A correlation between residuals therefore rewards shrinkage towards zero and should not be used alone to claim GI prediction; a variance-explained measure relative to the additive prediction avoids this.

3.5 Adamson CRISPRi screen

On the Adamson data, which contain only single perturbations, TBD (Table ??).

4 Discussion

We set out to test whether conditioning a linear-time state-space model on a gene–gene graph yields better predictions of perturbation responses than simple baselines. With inputs that represent every perturbation, a training procedure that starts from the train-mean prediction, and controls that isolate each component, the answer is mixed. The graph helped where the model has least to go on—single perturbations of genes never perturbed in training—and a permuted graph did not, so the model does extract information from co-expression neighbourhoods. The improvement was, however, small and comparable to the variation between splits. For double perturbations, the sum of single effects remained the best predictor, consistent with Ahlmann-Eltze et al. [2025], and none of our models, nor GEARS, explained any of the variance of genetic interactions beyond it.

The mechanism that motivated the architecture—modulating the SSM step size by graph proximity—had no measurable effect in any of our experiments, whether the conditioning weights started at zero or at random values. The benefit on unseen singles came from supplying the graph signal as a token feature, which a non-recurrent model can also exploit. We see two reasons. First, a step-size change alters how quickly the state forgets, but with a bidirectional scan over 5,045 tokens and short effective memory, most of the information a token needs is already available locally or through the shared perturbation embedding. Second, the Norman responses are dominated by a few programs (e.g. erythroid and granulocyte differentiation in K562), so a low-rank readout captures

most of the explainable variance, leaving little that a gene-specific propagation mechanism could add.

Our results carry three practical lessons. (i) The input encoding must be audited: an encoding tied to the output gene set discarded most perturbations in an earlier version of this work without any visible symptom in training. (ii) Deep models for this task should be initialised at, or explicitly compared with, the train-mean and additive predictors; without that initialisation our model did not improve on the train mean at all. (iii) Genetic-interaction prediction should be measured as variance explained relative to the additive prediction on doubles whose singles were observed; correlations between residuals reward shrinkage.

Limitations. We evaluated two K562 screens with the standard GEARS splits; screens in other cell types, genome-wide single-perturbation screens and larger combinatorial libraries may behave differently, and the Replogle et al. screens could not be included because most of their targets are not among the measured genes in the GEARS release, which our token-based encoding requires. Each split was trained with one seed, so between-split variation includes model-initialisation variation. Hyperparameters were chosen on the validation set of the first split only, and not tuned per variant. We used a single co-expression graph from control cells; knowledge graphs such as Gene Ontology or protein interaction networks [Roohani et al., 2024, Wenkel et al., 2026] might carry complementary information. Finally, our PyTorch scan runs on CPU; a fused GPU kernel would change absolute run times but not the linear scaling.

Acknowledgements

We thank the authors of GEARS for releasing their code and processed data.

Author contributions

A.P. designed the model and performed the computational experiments; R.H. contributed to the biological interpretation and to writing. Both authors approved the final manuscript.

Funding

This work received no specific funding.

Conflict of interest: none declared.

Data availability

The Norman et al. and Adamson et al. data are available from NCBI GEO (GSE133344 and GSE90546) and in processed form from the GEARS

repository. Code, split definitions, per-perturbation metrics and scripts that regenerate every table and figure are available at <https://github.com/The-MathKing/gcpmamba>; predictions of all models will be deposited in Zenodo.

References

- Britt Adamson, Thomas M. Norman, Marco Jost, et al. A multiplexed single-cell CRISPR screening platform enables systematic dissection of the unfolded protein response. *Cell*, 167(7):1867–1882, 2016.
- Constantin Ahlmann-Eltze, Wolfgang Huber, and Simon Anders. Deep-learning-based gene perturbation effect prediction does not yet outperform simple linear baselines. *Nature Methods*, 22:1657–1661, 2025.
- Gerold Csentes, Gema Sanz, Kristóf Z. Szalay, and Bénce Szalai. Benchmarking foundation cell models for post-perturbation RNA-seq prediction. *BMC Genomics*, 26:393, 2025.
- Haotian Cui, Chloe Wang, Hassaan Maan, Kuan Pang, Fengning Luo, Nan Duan, and Bo Wang. scGPT: toward building a foundation model for single-cell multi-omics using generative AI. *Nature Methods*, 21: 1470–1480, 2024.
- Atrey Dixit, Oren Parnas, Biyu Li, et al. Perturb-Seq: dissecting molecular circuits with scalable single-cell RNA profiling of pooled genetic screens. *Cell*, 167(7):1853–1866, 2016.
- Miroslav Fiedler. Algebraic connectivity of graphs. *Czechoslovak Mathematical Journal*, 23(2):298–305, 1973.
- Albert Gu and Tri Dao. Mamba: Linear-time sequence modeling with selective state spaces. *arXiv preprint arXiv:2312.00752*, 2023.
- Ilya Loshchilov and Frank Hutter. Decoupled weight decay regularization. In *International Conference on Learning Representations*, 2019.
- Thomas M. Norman, Max A. Horlbeck, Joseph M. Replogle, Alex Y. Ge, Albert Xu, Marco Jost, Luke A. Gilbert, and Jonathan S. Weissman. Exploring genetic interaction manifolds constructed from rich single-cell phenotypes. *Science*, 365(6455):786–793, 2019.
- Joseph M. Replogle, Reuben A. Saunders, Angela N. Pogson, et al. Mapping information-rich genotype-phenotype landscapes with genome-scale Perturb-seq. *Cell*, 185(14):2559–2575, 2022.
- Yusuf Roohani, Kexin Huang, and Jure Leskovec. Predicting transcriptional outcomes of novel multigene perturbations with GEARS. *Nature Biotechnology*, 42:927–935, 2024.
- Christina V. Theodoris, Ling Xiao, Anant Chopra, et al. Transfer learning enables predictions in network biology. *Nature*, 618:616–624, 2023.
- Ashish Vaswani, Noam Shazeer, Niki Parmar, et al. Attention is all you need. In *Advances in Neural Information Processing Systems*, volume 30, 2017.
- Frederik Wenkel et al. TxPert: using multiple knowledge graphs for prediction of transcriptomic perturbation effects. *Nature Biotechnology*, 2026. doi:10.1038/s41587-026-03113-4.